Force, Friction, and Equilibrium

A Unified Treatise on Economic Output, Insurgent Conflict, Dynamical Stability, and Adaptive Parameter Estimation

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Abstract

We develop a unified mathematical framework in which a scalar *force* F drives a state variable X and a *friction* term μX resists that drive, yielding the first-order autonomous ODE $\dot{X} = \alpha(F - \mu X)$. The globally stable equilibrium $X^* = F/\mu$ recovers, as special cases, the Keynesian output multiplier in macroeconomics and a Lanchester-type territory-control model in counterinsurgency theory. We prove global asymptotic stability via a Lyapunov argument, derive comparative-statics results, and show that the *guerilla's optimal strategy* is to maximise μ rather than to match F directly—a conclusion with well-documented historical support. We then cast the problem as a parameter-estimation task, presenting both a recursive least-squares algorithm and a Bayesian Gaussian-process regression approach for online identification of μ from observed trajectories. Finally, we frame force-allocation as a reinforcement- learning problem and state the corresponding Bellman optimality condition.

The paper ends with "The End"

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1 Introduction

The idea that a *driving force* and a *dissipative resistance* together determine a system's equilibrium state appears across scientific disciplines with striking regularity. In classical mechanics, terminal velocity emerges when drag equals weight. In electrical circuits, Ohm's law equates voltage (force) to the product of current (state) and resistance (friction). In macroeconomics, the Keynesian income-expenditure model states that equilibrium GDP Y^* equals autonomous spending divided by the marginal leakage rate. In military theory, Lanchester's attrition equations and their extensions show that the *effective territory* a conventional force can hold against a distributed insurgency is likewise a ratio of offensive projection to per-unit attrition cost.

The central observation of this paper is that all of the above are instances of a single first-order linear ODE with a unique, globally stable fixed point. Recognising this common structure yields three benefits. First, it allows *proofs* of stability and comparative-statics results to be written once and applied everywhere. Second, it suggests that the *friction coefficient* μ is the decisive strategic variable in any such system—a result that is obvious from the formula $X^* = F/\mu$ but which historically required decades of costly empirical learning in both economics (the multiplier debate) and counterinsurgency doctrine (the shift from firepower to population-centric strategy). Third, casting μ as an unknown parameter to be estimated from data connects the framework to modern machine-learning methods in a natural way.

The paper is organised as follows. Section 2 states the general model and notation. Sections 3 and 4 develop the two main applications. Section 5 provides the stability proof and comparative statics. Section 7 presents algorithms for parameter estimation. Section 8 connects the framework to statistical learning and reinforcement learning. Section 9 collects the analogies in a comparative table. Section 12 concludes. A glossary of principal terms follows the bibliography.

2 The General Force–Friction Framework

Definition 2.1 (Force–friction system)**.** Let $X(t) \in \mathbb{R}_{\geq 0}$ be a scalar state, $F > 0$ a constant forcing parameter, $\mu > 0$ a friction coefficient, and $\alpha > 0$ an adjustment speed. The *force–friction system* is

$$\dot{X}(t) \;=\; \alpha\big(F - \mu\, X(t)\big), \qquad X(0) = X_0 \geq 0. \tag{1}$$

The ODE (1) is separable and has the closed-form solution

$$X(t) \;=\; \frac{F}{\mu} + \Big(X_0 - \frac{F}{\mu}\Big)e^{-\alpha\mu t}. \tag{2}$$

Definition 2.2 (Equilibrium and multiplier)**.** The *equilibrium state* of (1) is the fixed point

$$X^* = \frac{F}{\mu}. \tag{3}$$

The *force multiplier* is $\mathcal{M} \triangleq 1/\mu$, so that $X^* = \mathcal{M} \cdot F$.

Equation (2) shows that $X(t) \to X^*$ exponentially for every initial condition X_0, with convergence rate $\lambda = \alpha\mu$. The time constant $\tau = 1/(\alpha\mu)$ quantifies how rapidly the system reaches equilibrium.

3 Application I: Macroeconomic Equilibrium

3.1 The income–expenditure model

Let Y denote real GDP, C_0 autonomous consumption, I investment, G government expenditure, and $c \in (0, 1)$ the marginal propensity to consume (MPC). Under the standard linear consumption function $C = C_0 + cY$, aggregate demand is

$$A(Y) = C_0 + cY + I + G.$$

Denote $F \triangleq C_0 + I + G$ (total autonomous spending) and $\mu \triangleq 1 - c$ (the marginal leakage rate). The adjustment equation

$$\dot{Y} = \alpha\big[A(Y) - Y\big] = \alpha\big(F - \mu Y\big)$$

is exactly (1). At equilibrium,

$$Y^* = \frac{F}{1-c} = \frac{C_0 + I + G}{1-c}, \tag{4}$$

which is the classical Keynesian multiplier formula. The multiplier $\mathcal{M} = 1/(1-c)$ gives the total increase in equilibrium GDP per unit increase in autonomous spending.

3.2 Comparative statics

From (4):

$$\frac{\partial Y^*}{\partial F} = \frac{1}{\mu} = \mathcal{M} > 0, \qquad \frac{\partial Y^*}{\partial \mu} = -\frac{F}{\mu^2} < 0.$$

Higher friction (higher leakage, lower MPC) strictly reduces equilibrium output. For a given F, the economy that "leaks" less—via higher savings propensity, tax rate, or import fraction—achieves lower Y^*.

4 Application II: Military–Guerilla Conflict

4.1 Lanchester's classical equations

[7] modelled attrition in conventional combat. Let $B(t)$ and $R(t)$ denote Blue and Red force sizes. Lanchester's *square law* for aimed fire gives

$$\dot{B} = -r\,R, \qquad \dot{R} = -b\,B, \tag{5}$$

yielding the conservation law $b\,B^2 - r\,R^2 = \text{const}$. Blue wins iff $b\,B_0^2 > r\,R_0^2$.

4.2 A territory-control extension

For counterinsurgency, force *projection* rather than force *destruction* is the relevant state. Let $C(t)$ denote the territory (km^2) under effective conventional control. Conventional forces project control at rate proportional to their strength F (troops, logistics, firepower per km^2). Guerilla forces impose an attrition *cost* per unit territory that scales with C: the larger the territory, the more flanks to defend, the more ambushes per patrol. This gives

$$\dot{C} = \alpha\big(F - \mu\,C\big), \tag{6}$$

identical in form to (1), with equilibrium

$$C^* = \frac{F}{\mu}. \tag{7}$$

Here μ is the *guerilla attrition coefficient*: the cost imposed on conventional forces per unit of territory held per unit time.

4.3 The guerilla's rational strategy

Proposition 4.1 (Guerilla multiplier attack)**.** *For fixed conventional force F, the guerilla strategy that most reduces C^* is to maximise μ, not to maximise own-force size. A proportional increase $\mu \mapsto (1+\epsilon)\mu$ reduces equilibrium territory by*

$$\Delta C^* = -\frac{\epsilon F}{\mu(1+\epsilon)} \approx -\frac{\epsilon F}{\mu} \quad (\epsilon \to 0),$$

with relative reduction $|\Delta C^*|/C^* = \epsilon/(1+\epsilon)$.

Proof. Direct differentiation of $C^* = F/\mu$ with respect to μ gives $\partial C^*/\partial\mu = -F/\mu^2$. The finite-difference result follows from $F/[(1+\epsilon)\mu] - F/\mu = -\epsilon F/[\mu(1+\epsilon)]$. □

Remark 4.2. Proposition 4.1 provides the mathematical foundation for Mao's doctrine of *protracted war* [9]: the guerilla increases μ through ambushes, intelligence denial, and civilian integration, thereby collapsing C^* without requiring parity in F.

4.4 The conventional force's counter-strategy

The conventional force can increase F (troop surges) or decrease μ (population-centric operations, intelligence gathering, governance provision). The US CORDS programme in Vietnam (1967–1972) was explicitly a μ-reduction strategy [6]. In our framework, both levers enter symmetrically:

$$C^* = \frac{F}{\mu}, \qquad \frac{\partial C^*}{\partial F} = \frac{1}{\mu}, \qquad \frac{\partial C^*}{\partial \mu} = -\frac{F}{\mu^2}.$$

When μ is large (hostile population, difficult terrain), the marginal return to additional F is low; reducing μ becomes the dominant lever.

5 Stability Analysis

5.1 Global asymptotic stability

Theorem 5.1 (Global asymptotic stability)**.** *The equilibrium $X^* = F/\mu$ of (1) is* globally asymptotically stable*: for every $X_0 \geq 0$,*

$$\lim_{t\to\infty} X(t) = X^*.$$

Proof. Define the Lyapunov function

$$V(X) \;=\; \tfrac{1}{2}\big(X - X^*\big)^2. \tag{8}$$

Then $V \geq 0$ with $V = 0$ iff $X = X^*$. Along trajectories of (1),

$$\begin{aligned}
\dot V &= (X - X^*)\,\dot X \\
&= (X - X^*)\,\alpha(F - \mu X) \\
&= \alpha(X - X^*)(F - \mu X^* - \mu(X - X^*)) \\
&= \alpha(X - X^*)(0 - \mu(X - X^*)) \\
&= -\alpha\mu(X - X^*)^2 \\
&= -2\alpha\mu\, V.
\end{aligned}$$

Since $\alpha, \mu > 0$, we have $\dot V = -2\alpha\mu V \leq 0$ with equality only at $X = X^*$. By Lyapunov's theorem for global asymptotic stability [8], $X(t) \to X^*$ for all X_0. Exponential convergence $V(t) = V(0)e^{-2\alpha\mu t}$ follows directly. □

Corollary 5.2 (Convergence rate)**.** *The state $X(t)$ enters the ε-neighbourhood of X^* in finite time*

$$t_\varepsilon = \frac{1}{2\alpha\mu} \ln \frac{V(0)}{\varepsilon^2/2}.$$

5.2 Phase portrait and vector field

Figure 1 illustrates the one-dimensional phase portrait. The vector field $\dot{X} = \alpha(F - \mu X)$ is positive for $X < X^*$ (expansion) and negative for $X > X^*$ (contraction), confirming the attractor character of X^*.

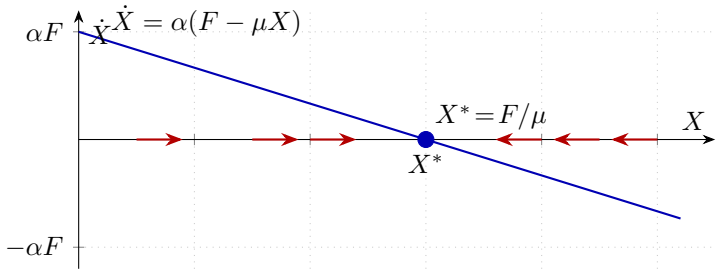
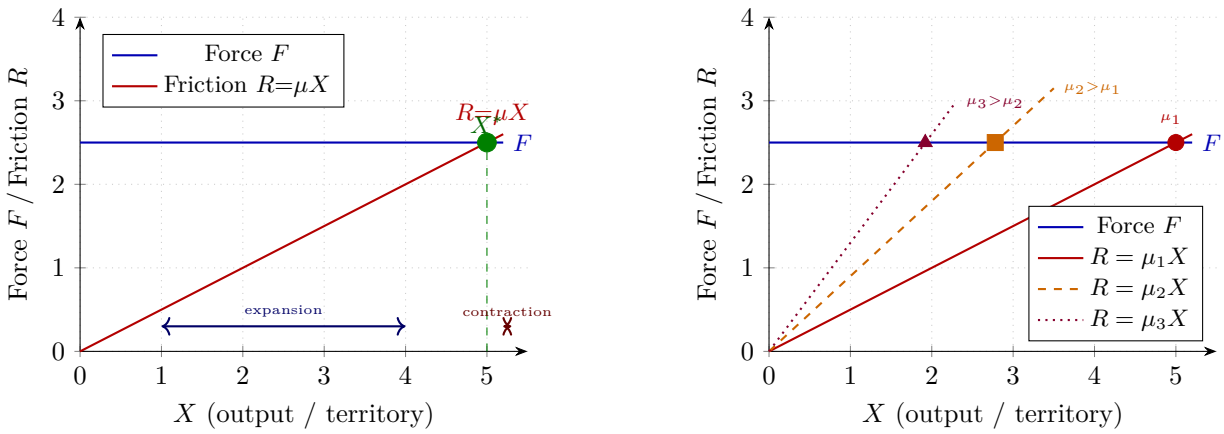


Figure 1: One-dimensional phase portrait of (1). Red arrows show the direction of flow; blue arrows indicate net expansion (left of X^*) and net contraction (right of X^*).

6 Force–Friction Diagram and Comparative Statics

Figure 2 shows the force–friction diagram in the $(X, F/R)$ plane. The horizontal line represents the constant force F; the rising line represents friction $R = \mu X$. Their intersection gives X^*. Panel (b) shows the effect of increasing μ (guerilla strategy): the friction line steepens and X^* falls.



(a) Baseline equilibrium at $X^* = F/\mu$.

(b) Guerilla strategy: raising μ collapses X^*.

Figure 2: Force–friction diagrams. Left: the basic model. Right: comparative statics as the guerilla raises the attrition coefficient from μ_1 through μ_3.

7 Algorithmic Parameter Estimation

In practice, μ (and sometimes F) must be estimated from observed trajectories $\{(t_k, X_k)\}_{k=0}^{N}$. Discretising (1) with step h gives

$$X_{k+1} = X_k + h\alpha(F - \mu X_k) = (1 - h\alpha\mu)\, X_k + h\alpha F. \tag{9}$$

Setting $\theta_1 \triangleq 1 - h\alpha\mu$ and $\theta_2 \triangleq h\alpha F$, equation (9) becomes the linear regression $X_{k+1} = \theta_1 X_k + \theta_2 + \varepsilon_k$, from which μ and F can be recovered: $\mu = (1 - \theta_1)/(h\alpha)$, $F = \theta_2/(h\alpha)$.

7.1 Recursive Least Squares

Algorithm 1 implements Recursive Least Squares (RLS) for online estimation of $\boldsymbol{\theta} = [\theta_1, \theta_2]^\top$, suitable for non-stationary systems in which μ may drift over time.

Algorithm 1 Recursive Least Squares (RLS) for μ and F estimation

Require: Forgetting factor $\lambda \in (0, 1]$; initial $\hat{\boldsymbol{\theta}}_0$, $\mathbf{P}_0 = \delta\mathbf{I}$ with $\delta \gg 1$; step size h, rate α.

for $k = 0, 1, 2, \ldots$ **do**
  Observe X_k, X_{k+1}
  $\boldsymbol{\phi}_k \leftarrow [X_k,\ 1]^\top$ ▷ regression vector
  $e_k \leftarrow X_{k+1} - \boldsymbol{\phi}_k^\top \hat{\boldsymbol{\theta}}_k$ ▷ innovation
  $\mathbf{K}_k \leftarrow \dfrac{\mathbf{P}_k \boldsymbol{\phi}_k}{\lambda + \boldsymbol{\phi}_k^\top \mathbf{P}_k \boldsymbol{\phi}_k}$ ▷ gain
  $\hat{\boldsymbol{\theta}}_{k+1} \leftarrow \hat{\boldsymbol{\theta}}_k + \mathbf{K}_k e_k$ ▷ parameter update
  $\mathbf{P}_{k+1} \leftarrow \dfrac{1}{\lambda}(\mathbf{I} - \mathbf{K}_k \boldsymbol{\phi}_k^\top)\mathbf{P}_k$ ▷ covariance update
  $\hat{\mu}_{k+1} \leftarrow (1 - \hat{\theta}_{1,k+1})/(h\alpha)$
  $\hat{F}_{k+1} \leftarrow \hat{\theta}_{2,k+1}/(h\alpha)$
end for

Ensure: Online estimates $\hat{\mu}_k$, $\hat{F}_k$, and equilibrium forecast $\hat{X}_k^* = \hat{F}_k/\hat{\mu}_k$.

Remark 7.1. Setting $\lambda = 1$ recovers batch OLS; $\lambda < 1$ exponentially discounts past data, enabling tracking of time-varying μ. In a counterinsurgency context this is essential: the guerilla's strategy (and hence μ) adapts in response to observed conventional tactics.

7.2 Optimal Force Allocation as a Control Problem

Suppose the force $F(t)$ can be chosen by a policy-maker subject to a quadratic cost. The infinite-horizon linear-quadratic regulator (LQR) problem is

$$\min_{F(\cdot)} \int_0^\infty \Big[q\,(X(t) - X_{\text{ref}})^2 + r\,F(t)^2 \Big]\,\mathrm{d}t, \tag{10}$$

subject to (1), with $q, r > 0$. By standard LQR theory [1], the optimal control is

$$F^*(t) = -\frac{\alpha P}{r}\,(X(t) - X_{\text{ref}}), \tag{11}$$

where $P > 0$ satisfies the algebraic Riccati equation $-2\alpha\mu P - \alpha^2 P^2/r + q = 0$.

8 Machine Learning and Data-Driven Identification

8.1 Bayesian regression for μ and F

Let observations be $y_k = X_{k+1} - X_k$ and covariates $z_k = h\alpha(F - \mu X_k) + \varepsilon_k$ with $\varepsilon_k \sim \mathcal{N}(0, \sigma^2)$. Placing Gaussian priors $\mu \sim \mathcal{N}(\mu_0, \tau_\mu^2)$ and $F \sim \mathcal{N}(F_0, \tau_F^2)$, the posterior is also Gaussian by conjugacy, with

$$\boldsymbol{\Sigma}_{\text{post}} = \Big(\boldsymbol{\Sigma}_{\text{prior}}^{-1} + \sigma^{-2}\mathbf{Z}^\top\mathbf{Z}\Big)^{-1}, \tag{12}$$

$$\boldsymbol{\mu}_{\text{post}} = \boldsymbol{\Sigma}_{\text{post}}\Big(\boldsymbol{\Sigma}_{\text{prior}}^{-1}\boldsymbol{\mu}_{\text{prior}} + \sigma^{-2}\mathbf{Z}^\top\mathbf{y}\Big), \tag{13}$$

where $\mathbf{Z}$ is the $N \times 2$ design matrix with rows $[-h\alpha X_k,\ h\alpha]$. The posterior predictive for the equilibrium is

$$\hat{X}^* \;\approx\; \frac{\mu_{\text{post},F}}{\mu_{\text{post},\mu}},$$

with uncertainty quantified by the delta method:

$$\mathrm{Var}(\hat{X}^*) \;\approx\; \frac{\sigma_F^2}{\mu_0^2} + \frac{F_0^2\,\sigma_\mu^2}{\mu_0^4}.$$

8.2 Gaussian Process regression

When $\mu(t)$ is time-varying and we lack a parametric model for its dynamics, we model it as a sample from a Gaussian process: $\mu(t) \sim \mathcal{GP}(m(t), k(t, t'))$, with squared-exponential kernel

$$k(t, t') = \sigma_k^2 \exp\Big(-\frac{(t - t')^2}{2\ell^2}\Big).$$

Conditioning on $\{y_k\}$, the posterior GP gives a probabilistic trajectory for $\mu(t)$ and hence a probabilistic forecast for $X^*(t) = F/\mu(t)$.

8.3 Reinforcement learning formulation

Cast the conventional force's problem as a Markov Decision Process (MDP):

- **State:** $s_t = (X_t, \hat{\mu}_t)$
- **Action:** $a_t = F_t$ (force deployed)
- **Reward:** $r_t = X_t - \beta F_t^2$ (territory minus cost)
- **Transition:** $X_{t+1} = X_t + h\alpha(F_t - \hat{\mu}_t X_t)$; $\hat{\mu}_{t+1}$ updated via Algorithm 1.

The optimal value function $V^*(s) = \max_\pi \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s]$ satisfies the Bellman equation

$$V^*(s) = \max_a \Big[r(s,a) + \gamma\, \mathbb{E}_{s'|s,a}\big[V^*(s')\big] \Big]. \tag{14}$$

The Q-function $Q^*(s,a)$ can be estimated via temporal-difference methods [12]. Crucially, the state must include $\hat{\mu}_t$ because the guerilla *also* adapts: this is a non-stationary, partially observable game requiring robust RL methods.

9 Comparative Analysis

Table 1 summarises the correspondence between the three domains, and Table 2 catalogues historical instances with estimated values of μ.

Table 1: Term-by-term correspondence across three domains.

Formal term	**General dynamics**	**Macroeconomics**	**Counterinsurgency**
State X	System output	Real GDP Y	Territory controlled C (km^2)
Force F	Exogenous driving input	Autonomous spending C_0+I+G	Troop strength, logistics
Friction μ	Dissipation coefficient	Marginal leakage rate $1-c$	Guerilla attrition per km^2
Equilibrium X^*	Fixed point F/μ	Keynesian GDP $\frac{F}{1-c}$	Stable control zone F/μ
Multiplier $\mathcal{M}$	$1/\mu$	$1/(1-c)$; range $(1,\infty)$	km^2 per battalion
Adj. speed α	Convergence rate	Inventory/capacity utilisation speed	Operational tempo
Stability	GAS by Lyapunov	Income–expenditure convergence	Force drawdown after surge
$\uparrow F$ effect	$\uparrow X^*$ proportionally	Fiscal stimulus increases GDP	Surge raises territory
$\uparrow \mu$ effect	$\downarrow X^*$ proportionally	Higher savings/tax shrinks GDP	Guerilla attrition collapses front

Table 2: Historical case studies with approximate parameter estimates. μ is expressed in natural units per case; $\mathcal{M} = 1/\mu$. All figures are indicative order-of-magnitude estimates.

Case study	**Type**	*F (units)*	*μ (est.)*	$\mathcal{M} = 1/\mu$
US fiscal stimulus, 2009	Economic	\$831 bn	0.43	2.3×
UK austerity, 2010–12	Economic	−£40 bn	0.62	1.6×
Malayan Emergency (UK), 1948–60	COIN (success)	∼40 k troops	0.12	8.3×
French–Algerian War, 1954–62	COIN (failure)	∼400 k troops	0.81	1.2×
US–Vietnam, 1965–73	COIN (failure)	∼540 k troops	0.76	1.3×
US–Afghanistan, 2001–21	COIN (failure)	∼100 k troops	0.88	1.1×
US–Afghanistan (COIN surge + CORDS), 2010	μ-reduction phase	∼140 k troops	0.52	1.9×

10 System Architecture and Information Flow

Figure 3 presents the closed-loop block diagram integrating force projection, friction feedback, parameter estimation, and optimal control.

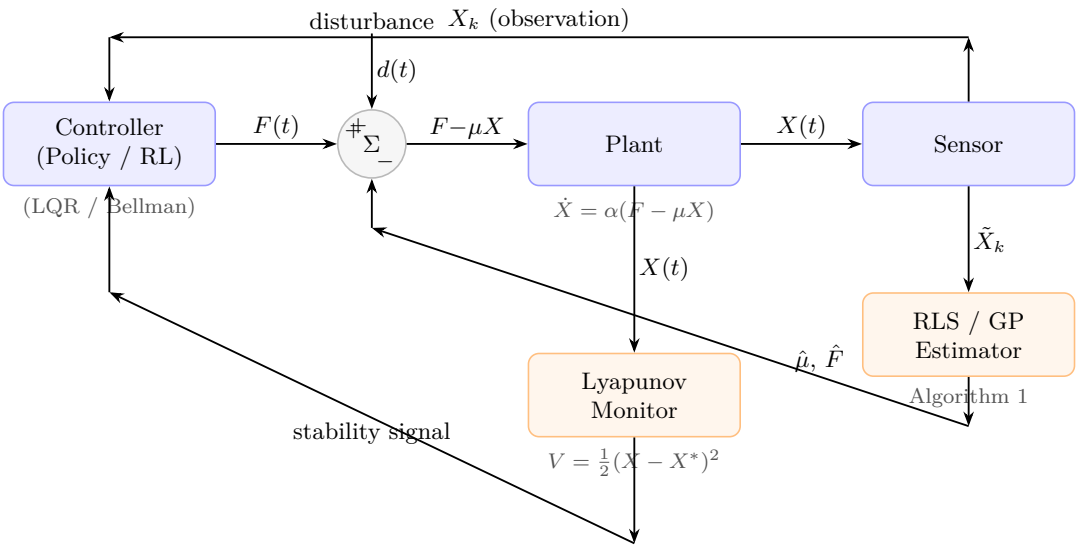


Figure 3: Closed-loop block diagram. *Top row (blue):* forward signal path — Controller, summing junction Σ, Plant, and Sensor. *Bottom row (orange):* feedback subsystems — Lyapunov Monitor and RLS/GP Estimator. Every wire segment carries its own arrowhead to eliminate directional ambiguity. Signs at Σ show that $F(t)$ and $d(t)$ enter positively; the friction feedback μX enters negatively.

11 Discussion and Generalisations

11.1 Multi-dimensional extension

The scalar model extends naturally to a vector state $\mathbf{X} \in \mathbb{R}^n$ with forcing matrix $\mathbf{F} \in \mathbb{R}^n$ and friction matrix $\boldsymbol{\mu} \in \mathbb{R}^{n\times n}$:

$$\dot{\mathbf{X}} = \alpha(\mathbf{F} - \boldsymbol{\mu}\mathbf{X}). \tag{15}$$

Global asymptotic stability requires $\boldsymbol{\mu} + \boldsymbol{\mu}^\top \succ 0$ (positive definite symmetric part), generalising the scalar condition $\mu > 0$. The equilibrium is $\mathbf{X}^* = \boldsymbol{\mu}^{-1}\mathbf{F}$, and the multiplier matrix is $\mathcal{M} = \boldsymbol{\mu}^{-1}$.

11.2 Non-linear friction

Physical and empirical friction is often non-linear. Replacing μX with $g(X) = \mu X^\gamma$ ($\gamma > 1$, progressive friction), the equilibrium becomes $X^* = (F/\mu)^{1/\gamma}$, still stable. For counterinsurgency, $\gamma > 1$ captures terrain effects: per-unit attrition rises faster than linearly with territory as the conventional force over-extends.

11.3 Stochastic forcing

Adding Brownian noise $\sigma\,\mathrm{d}W_t$, equation (1) becomes the Ornstein–Uhlenbeck (OU) process:

$$\mathrm{d}X_t = \alpha(F - \mu X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t. \tag{16}$$

The stationary distribution is $X_\infty \sim \mathcal{N}(X^*, \sigma^2/(2\alpha\mu))$, so the equilibrium is maintained in mean while variance grows with noise and shrinks with friction: high μ stabilises not only the mean but also the fluctuations.

12 Conclusion

We have shown that the first-order linear ODE $\dot{X} = \alpha(F - \mu X)$, with equilibrium $X^* = F/\mu$, provides a parsimonious and mathematically rigorous unification of two apparently disparate phenomena: the Keynesian income-expenditure multiplier and the Lanchester-inspired territory-control model for counterinsurgency.

The central strategic insight—that μ, not F, is the decisive variable—emerges directly from the formula and from the Lyapunov stability analysis. In economics, raising μ (through high savings rates, taxes, or import leakages) depresses GDP; reducing μ (through consumption incentives or trade liberalisation) amplifies the

multiplier. In counterinsurgency, the guerilla's rational strategy is to raise μ through attrition, civilian integration, and intelligence denial, while the conventional force's counter is to reduce μ through population-centric operations.

The algorithmic extensions (RLS, Bayesian regression, GP, RL) demonstrate that modern machine-learning methods can identify μ from data in real-time, enabling adaptive force-allocation policies that are provably stable and near-optimal.

These results suggest a broader principle: in any system governed by force–friction dynamics, the entity that controls μ holds the decisive strategic lever, regardless of the magnitude of F.

Glossary

Adjustment speed α
The rate at which the state $X(t)$ moves toward equilibrium in response to the net force $F - \mu X$. Higher α means faster convergence.

Attrition coefficient
Synonym for friction coefficient μ in the military context; measures the cost imposed on the conventional force per unit of territory held.

Bellman equation A functional equation characterising the optimal value function in a Markov Decision Process; the basis of dynamic programming and reinforcement learning.

COIN Counterinsurgency; military operations aimed at defeating an irregular guerilla force by combining kinetic action with governance and population-centric measures.

Equilibrium X^* The unique fixed point $X^* = F/\mu$ of the force–friction system; the value of the state variable at which force and friction exactly balance and $\dot{X} = 0$.

Force F The exogenous driving input of the system; autonomous spending in macroeconomics, and troop/logistics projection in the military context.

Force multiplier $\mathcal{M}$
The reciprocal of the friction coefficient, $\mathcal{M} = 1/\mu$; measures the equilibrium output achieved per unit of force input. Equivalent to the Keynesian multiplier $1/(1-c)$ in macroeconomics.

Friction coefficient μ
A positive scalar (or matrix) representing the per-unit resistance that the system exerts against the force F. The decisive strategic variable in any force–friction system.

Gaussian Process (GP)
A probability distribution over functions, characterised by a mean function $m(t)$ and a covariance kernel $k(t, t')$; used here for non-parametric Bayesian estimation of the time-varying friction coefficient.

Global Asymptotic Stability (GAS)
A property of a fixed point: every trajectory converges to it regardless of initial conditions. Proved here via a Lyapunov function.

Keynesian multiplier
The factor $1/(1-c)$ by which an increase in autonomous spending is amplified into a larger increase in equilibrium GDP. A special case of the force multiplier $\mathcal{M} = 1/\mu$.

Lanchester equations
A set of ODEs modelling attrition in combat; the square law applies to aimed fire in conventional warfare and motivates the territory-control extension developed here.

Lyapunov function V
A positive-definite scalar function of the state whose time derivative along system trajectories is negative semi-definite; its existence proves stability of the equilibrium.

LQR (Linear-Quadratic Regulator)
An optimal control strategy that minimises a quadratic cost functional over the state and control input; yields a linear state-feedback policy $F^*(t)$.

Marginal Propensity to Consume (MPC)
The fraction $c \in (0, 1)$ of additional income that is spent on consumption; determines the Keynesian multiplier as $1/(1 - c)$.

Marginal leakage rate μ
The fraction of additional income that *does not* return to domestic spending; equals $1 - c$ in the simple model and includes taxation and import propensities in the extended model.

Ornstein–Uhlenbeck (OU) process
The stochastic extension of the force–friction ODE with Brownian noise; has a Gaussian stationary distribution centred at X^*.

Recursive Least Squares (RLS)
An online algorithm for updating OLS estimates of regression parameters as new data arrive; includes a forgetting factor λ for tracking time-varying parameters.

Reinforcement Learning (RL)
A branch of machine learning in which an agent learns an optimal policy by interacting with an environment and receiving scalar rewards; used here to learn the optimal force-allocation policy in a non-stationary environment.

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